

8_popquiz review test -

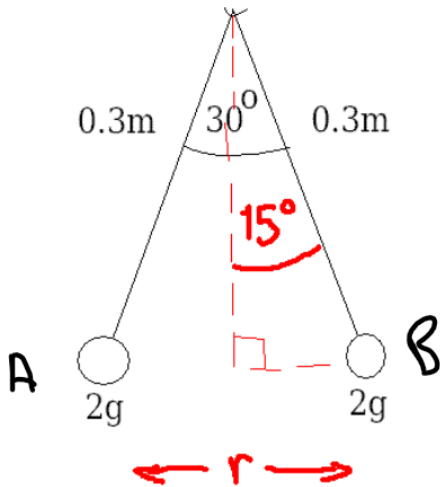
⚠ This is a preview of the published version of the quiz

Started: Mar 15 at 3:12pm

Quiz Instructions



Question 1 25 pts



consider 2 charges ($q_1=q_2=a$) hanging on strings. they are (obviously) repelling each other and the angle between the strings is 30 degrees. I will break down the problem for you.

First find the distance d between the charges.

☐ 0.0775m

☐ 0.58m

☐ 0.155m

☐ 0.3m



Question 2 25 pts

Building on the previous question. At location B, the electric field is $1.9 \cdot 10^6 \text{ N/C}$



is 0



IDK



is @ right



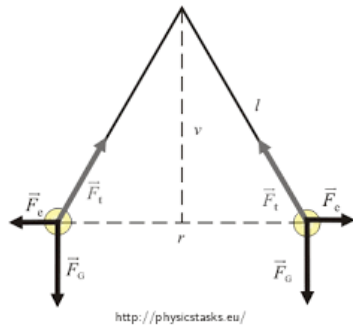
is @left



Question 3 25 pts

Building on the previous question. the electric field is $1.9 \cdot 10^6$ @ right N/C

the goal is to find q . Focus on B . It is acted upon by 3 forces . weight (mg), tension T and the electric field qE . I help you . here is the free-body diagram.



Its in equilibrium so :

Equ1: $T_x = qE$ (along x) and Equ2: $T_y = mg$ (along y). the mass is $2g = 0.002kg$.

Can you solve for q ?

hint: $T_x = T \sin(\text{angle})$ of course. $T_y = T \cos(\text{angle})$

Then do: (Equ1)/(Equ2) -> that will eliminate T . solve for q . that easy ($\sin/\cos = \tan$)

nC is nano uC is micro



about 5 uC



about 0.3 nC



about 2 nC



about 3 nC



Question 4 25 pts

The units for electric flux is

☐

N C /m

☐

$(\text{N m}^2)/\text{C}$

☐

m^2/C

☐

N m

Not saved

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